

Generation Interconnection System Impact Study Report

For Queue Project Q18-0995

North Creek 138 kV

May 17, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0995. The Interconnection Customer proposes a 211.3 MW solar photovoltaic facility interconnecting at the North Creek 138 kV substation in the Kentucky area. The request was received on May 13, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 415 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$76,555,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: N. SHANNON to N CREEK 138 kV
Contingency: MULLBERRY SUBSTATION to SALEM WALK 138 kV line
Loading: 168.6% (Rating: 231 MVA)
DFAX: 79.1%
Pre-project loading: 96.2%
2. Facility: REYNOLDS HILL to FORT RUCKER 138 kV Line
Contingency: NORTH CRK to N. SHANNON SUBSTATION 138 kV line
Loading: 149.0% (Rating: 257 MVA)
DFAX: 100.0%
Pre-project loading: 66.7%
3. Facility: HARE to DLUZERNE 138 kV
Contingency: WEBER to LARK ELLEN 138 kV
Loading: 138.4% (Rating: 195 MVA)
DFAX: 38.2%
Pre-project loading: 97.0%
4. Facility: ELBERFELD SOUTH to STUDIO 138 kV
Contingency: HARE to N. SHANNON 138 kV
Loading: 126.6% (Rating: 223 MVA)
DFAX: 38.8%
Pre-project loading: 89.7%
5. Facility: N. SHANNON to HARE 138 kV L
Contingency: WEBER to LARK ELLEN 138 kV
Loading: 125.3% (Rating: 343 MVA)
DFAX: 73.1%
Pre-project loading: 80.3%
6. Facility: POTATO RAPIDS to PEACOCK 138 kV
Contingency: N. SHANNON SUBSTATION to HARE SUBSTATION 138 kV
Loading: 123.1% (Rating: 243 MVA)

DFAX: 35.2%
Pre-project loading: 92.5%

7. Facility: HILL COUNTRY 345 to PEACOCK 138 kV L
Contingency: N. SHANNON to HARE 138 kV L
Loading: 121.0% (Rating: 252 MVA)
DFAX: 35.2%
Pre-project loading: 91.5%
8. Facility: POTATO RAPIDS to REC BRADFORD 138 kV
Contingency: HARE to N. SHANNON 138 kV
Loading: 120.8% (Rating: 228 MVA)
DFAX: 35.2%
Pre-project loading: 88.0%
9. Facility: EDNA to SUMMIT LAKE WPS 138 kV
Contingency: LARK ELLEN to WEBER 138 kV
Loading: 120.1% (Rating: 223 MVA)
DFAX: 33.1%
Pre-project loading: 88.8%
10. Facility: LAGUA TAP to STUDIO SUBSTATION 138 kV line
Contingency: N. SHANNON to HARE SUBSTATION 138 kV line
Loading: 120.1% (Rating: 343 MVA)
DFAX: 64.8%
Pre-project loading: 80.2%
11. Facility: LAGUNA TAP SUBSTATION to LARK ELLEN 138 kV
Contingency: HARE to N. SHANNON SUBSTATION 138 kV
Loading: 119.6% (Rating: 349 MVA)
DFAX: 64.8%
Pre-project loading: 80.5%
12. Facility: REC BRADFORD to CURRY COUNTY 138 kV
Contingency: N. SHANNON SUBSTATION to HARE 138 kV Line
Loading: 119.5% (Rating: 215 MVA)
DFAX: 35.2%
Pre-project loading: 84.9%
13. Facility: WEBER to LARK ELLEN 138 kV Line
Contingency: N. SHANNON to HARE 138 kV line
Loading: 119.3% (Rating: 354 MVA)
DFAX: 64.8%
Pre-project loading: 80.7%
14. Facility: ABAJO SUBSTATION to FORT RUCKER 138 kV
Contingency: N. SHANNON to HARE SUBSTATION 138 kV
Loading: 118.7% (Rating: 274 MVA)
DFAX: 35.2%
Pre-project loading: 91.6%
15. Facility: SHEFFIELD to CURRY COUNTY 138 kV
Contingency: N. SHANNON to HARE 138 kV line
Loading: 118.2% (Rating: 213 MVA)
DFAX: 35.2%
Pre-project loading: 83.3%
16. Facility: SUMMIT LAKE WPS to SUGAR CRK POWER SUBSTATION 138 kV
Contingency: WEBER to LARK ELLEN 138 kV
Loading: 117.9% (Rating: 216 MVA)
DFAX: 33.1%

Pre-project loading: 85.6%

17. Facility: SHEFFIELD to JACINTO 138 kV L

Contingency: N. SHANNON to HARE 138 kV

Loading: 116.6% (Rating: 210 MVA)

DFAX: 35.2%

Pre-project loading: 81.2%

18. Facility: JACINTO to PG&E; PUMP SUBSTATION 138 kV Line

Contingency: HARE to N. SHANNON 138 kV L

Loading: 115.5% (Rating: 207 MVA)

DFAX: 35.2%

Pre-project loading: 79.6%

19. Facility: PG&E; PUMP to ALAMEDA 138 kV

Contingency: N. SHANNON SUBSTATION to HARE 138 kV

Loading: 115.0% (Rating: 198 MVA)

DFAX: 35.2%

Pre-project loading: 77.4%

20. Facility: BEAUMEADE to WEBER 138 kV

Contingency: HARE to N. SHANNON 138 kV Line

Loading: 113.0% (Rating: 194 MVA)

DFAX: 56.7%

Pre-project loading: 51.4%

21. Facility: BOONVILLE to SUGAR CREEK POWER 138 kV

Contingency: FT RUCKER to ABAJO 138 kV

Loading: 105.8% (Rating: 182 MVA)

DFAX: 26.6%

Pre-project loading: 74.9%

22. Facility: WEBER to MULLBERRY 138 kV Line

Contingency: N. SHANNON SUBSTATION to NORTH CRK SUBSTATION 138 kV

Loading: 105.3% (Rating: 363 MVA)

DFAX: 62.4%

Pre-project loading: 69.0%

23. Facility: MULLBERRY to SALEM WALK 138 kV

Contingency: N CRK to N. SHANNON 138 kV

Loading: 105.1% (Rating: 371 MVA)

DFAX: 62.4%

Pre-project loading: 69.6%

24. Facility: ROCKWOOD to STONY HILL 138 kV

Contingency: N. SHANNON to HARE 138 kV

Loading: 103.8% (Rating: 243 MVA)

DFAX: 38.0%

Pre-project loading: 70.7%

25. Facility: ROCKWOOD to W JAX 138 kV L

Contingency: HARE to N. SHANNON 138 kV

Loading: 103.4% (Rating: 267 MVA)

DFAX: 38.0%

Pre-project loading: 73.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N. SHANNON to N CRK 138 kV	Rebuild line with higher-capacity conductor	\$10,067,000	43% (\$4,366,000)
REYNOLDS HILL to FORT RUCKER 138 kV	Rebuild line with higher-capacity conductor	\$11,846,000	38% (\$4,451,000)
HARE to DLUZERNE 138 kV	Rebuild line with higher-capacity conductor	\$9,814,000	41% (\$4,025,000)
ELBERFELD S to STUDIO 138 kV	Rebuild line with higher-capacity conductor	\$8,872,000	40% (\$3,519,000)
N. SHANNON to HARE 138 kV L	Rebuild line with higher-capacity conductor	\$6,228,000	41% (\$2,554,000)
POTATO RAPIDS to PEACOCK 138 kV	Rebuild line with higher-capacity conductor	\$10,241,000	34% (\$3,519,000)
HILL COUNTRY 345 to PEACOCK 138 kV L	Rebuild line with higher-capacity conductor	\$6,158,000	34% (\$2,116,000)
POTATO RAPIDS to REC BRADFORD 138 kV	Rebuild line with higher-capacity conductor	\$6,536,000	34% (\$2,246,000)
EDNA to SUMMIT LAKE WPS 138 kV	Rebuild line with higher-capacity conductor	\$5,415,000	41% (\$2,222,000)
STUDIO to LAGUNA TAP 138 kV	Rebuild line with higher-capacity conductor	\$5,768,000	40% (\$2,284,000)
LAGUNA TAP SUBSTATION to LARK ELLEN 138 kV	Rebuild line with higher-capacity conductor	\$4,268,000	40% (\$1,690,000)
REC BRADFORD to CURRY COUNTY 138 kV	Rebuild line with higher-capacity conductor	\$6,440,000	34% (\$2,213,000)
WEBER to LARK ELLEN 138 kV Line	Rebuild line with higher-capacity conductor	\$14,681,000	40% (\$5,812,000)
ABAJO SUBSTATION to FORT RUCKER 138 kV	Rebuild line with higher-capacity conductor	\$5,726,000	34% (\$1,968,000)
SHEFFIELD to CURRY COUNTY 138 kV	Rebuild line with higher-capacity conductor	\$5,424,000	34% (\$1,864,000)
SUMMIT LAKE WPS to SUGAR CRK POWER SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$10,880,000	41% (\$4,465,000)
SHEFFIELD to JACINTO 138 kV L	Rebuild line with higher-capacity conductor	\$3,057,000	34% (\$1,050,000)
JACINTO to PG&E; PUMP 138 kV	Rebuild line with higher-capacity conductor	\$3,232,000	34% (\$1,111,000)
PG&E; PUMP to ALAMEDA 138 kV	Rebuild line with higher-capacity conductor	\$10,809,000	34% (\$3,714,000)
BEAUMEADE to WEBER 138 kV L	Rebuild line with higher-capacity conductor	\$31,104,000	58% (\$18,050,000)
BOONVILLE to SUGAR CREEK POWER 138 kV	Reconductor line	\$1,860,000	35% (\$654,000)
WEBER to MULLBERRY 138 kV	Reconductor line	\$1,088,000	38% (\$409,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MULLBERRY to SALEM WALK 138 kV	Reconductor line	\$1,879,000	38% (\$706,000)
ROCKWOOD to STONY HILL 138 kV	Reconductor line	\$1,131,000	40% (\$449,000)
ROCKWOOD to W JAX 138 kV L	Reconductor line	\$2,767,000	40% (\$1,098,000)
	Total allocated to project		\$76,555,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.